Fengjie Hong

Third-year Undergraduate Student

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Summary

Third-year undergraduate at the Institute for Interdisciplinary Information Sciences (IIIS), Tsinghua University, with a strong background in physics. Currently working on diffusion language models and reinforcement learning.

Education

Tsinghua University Beijing, China
Institute for Interdisciplinary Information Sciences (IIIS) Sep. 2024 – June 2028 (Expected)
GPA: 3.9/4.0; Rank: 8/90 (Top 9%)

Research Interests

Broad interests in artificial intelligence, with a current focus on diffusion language models and reinforcement learning. Open to exploring new directions across AI.

Research Experience

Research Intern, Impossible Research July 2026 – Present

Undergraduate Research Present

- Working with Haozhe Jiang in Prof. Nika Haghtalab's group at UC Berkeley on diffusion language models and reinforcement learning.
- Participate in Prof. Kaifeng Lyu's group meetings at Tsinghua University.

Technical Skills

- **Programming Languages**: Python, C++
- **Frameworks & Tools**: PyTorch, Triton, NumPy, Git, LaTeX, etc.
- **Languages**: Chinese (Native), English

Academic Projects

I have not yet published papers at academic conferences or in academic journals.

DiPOD Coding | *Ongoing research project* Autumn 2026

- Follow-up of DiPOD. Applied self-distillation to the RL of diffusion language models.
- Developed new technical designs to adapt DiPOD to a newly released diffusion languange model.
- Empirically validated DiPOD's theoretical predictions.

VideoEditor | *Course: Deep Learning* Spring 2026

- Contributed to a video object editing system supporting text prompts, reference images, and sketches through FLUX/AnyDoor first-frame editing and VideoPainter/CogVideoX propagation.
- The system combines sketch-to-image generation and object-mask tracking with React and Gradio interfaces for interactive editing.

Emotion Visualization | *Course: Computer Vision* Spring 2026

- Built a facial emotion recognition pipeline combining RetinaFace detection with a pretrained ViT classifier, supporting seven emotion categories and per-face attention visualization.
- Implemented enhanced CLS attention maps and integrated training-free test-time register tokens to reduce attention artifacts, with an interactive web interface and batch inference.

CVPR | *Course: Type-safe Modern System Practice* Autumn 2025

- We present Centralized Visual Package Router (CVPR), a typesafe front-end and back-end system for visualized package management.
- We build the system with functional programming paradigms and type-safe design principles.

Realtime Simulation | *Course: Advanced Computer Graphics* Autumn 2025

- We present a realtime fluid-rigid two way coupling and rendering system.
- We use PBD style method for simulation and NVIDIA Warp for multi-threading GPU acceleration.
- Achieved realtime movement control.

ME | *Self-Motivated* Autumn 2025

- To investigate differences between Marx and Engels, we select their writings and analyze semantic differences in key concepts using Word2Vec.
- Some differences can roughly be seen.

WAYF | *Course: Natural Language Processing* Autumn 2025

- In WAYF (Where Are You From), we present a classifier to detect translationese and fine-tune `Qwen3-4B` using the classifier to reduce translationese.
- For the classifier, we do transfer learning on BERT. We use GRPO to fine-tune `Qwen3-4B`, with the reward designed from the classifier's output.
- Our classifier outperforms zero-shot LLMs on translationese detection.

LABOTEX | *Course: Introduction to Large Language Model Applications* Spring 2025

- In LABOTEX, we present an agentic assistant designed to help researchers generate lab reports in LaTeX format easily and efficiently.
- We build 4 agents to read instruction books, write introduction section, process experiment data with tables and plots, and integrate all parts into a complete LaTeX report. These agents interact with each other through the intermediate files they generate.
- LABOTEX can generate nice LaTeX reports for General Physics' experiments.

Honors & Awards

- **Friends of Tsinghua - Lingjun Linghang Scholarship** 2026
- **Friends of Tsinghua - Weichen Scholarship** 2026
- **Tsinghua University Comprehensive Excellence Scholarship** 2026
- **Tsinghua University Merit Student** 2026
- **Friends of Tsinghua - Weichen Scholarship** 2025
- **Tsinghua University Comprehensive Excellence Scholarship** 2025
- **Freshman Scholarship** 2024
- **Gold Medal in 39th CPhO, 40th CPhO, 24th APhO** 2022-2024